

Pseudocode for an ABM of Brokerage in Matching Markets

1 State, Notation, and Matching Function

Blue notation, equations, and algorithm lines are diagnostics-only measures. They are recorded for $\mathcal{M}$ or downstream figures, exploration scripts, and analyses; they do not enter transition, choice, learning, or matching rules.

For any matrix Z , $Z_{\cdot\ell}$ denotes column ℓ and $Z_{\cdot L}$ denotes the submatrix with ordered column-index set L . For a vector z , z_L denotes the corresponding subvector.

Table 1: Cross-routine symbols and dimensions.

Symbol	Dimension/domain	Definition
<i>Core</i>		
N	scalar	Number of agents, with agent set $[N] = \{1, \dots, N\}$.
ϑ	parameter vector	Simulation parameter vector.
seed	integer	Random seed.
S^t	state tuple	Mutable simulation state at end of period t .
T	integer	Number of simulation periods.
$\mathcal{M} = [m^1; \dots; m^T]$	$\mathbb{R}^{T \times m}$	Measure table; row $m^t = \mathcal{M}_t \in \mathbb{R}^m$.
<i>Parameters</i>		
d	scalar	Agent type dimension.
k_G	scalar	Watts-Strogatz graph degree.
ω	scalar	Satisfaction EWMA weight.
E_{init}	scalar	Initial training steps.
E_{steps}	scalar	Minimum (floor) training steps per period.
η_{lr}	scalar	MLP Adam learning rate.
W_p	scalar	Training-window horizon, in periods.
M	scalar	Per-call training observation cap.
<i>Agent types, networks, and output</i>		
$G^t = (V, \mathcal{E}_G^t)$	graph	Undirected graph, $V = [N] \cup \{N + 1\}$; node $N + 1$ is the broker.
$X^t = [x_1^t, \dots, x_N^t]$	$\mathbb{R}^{d \times N}$	Agent type matrix; $x_i^t = X_{\cdot i}^t \in \mathbb{R}^d$.
γ	$[0, 1] \rightarrow \mathbb{R}^d$	Agent type curve for initial and entrant draws.
q_{ij}^t	scalar	Realized output for match (i, j) .
<i>True matching function</i>		
Q	scalar	Output offset.
ρ	scalar	Quality-interaction weight.
δ	scalar	Regime-gain amplitude.
σ_ε	scalar	Output-noise standard deviation.
c	$\mathbb{R}^d$	Ideal agent type.
A	$\mathbb{R}^{d \times d}$	Symmetric positive definite interaction matrix.
B	$\mathbb{R}^{d \times d}$	Symmetric regime operator.
<i>Calibration and costs</i>		

Symbol	Dimension/domain	Definition
$\bar{q}_{\text{cal}}$	scalar	Calibration mean.
r	scalar	Reservation output.
ϕ	scalar	Broker fee.
c_s	scalar	Self-search cost.
<i>Learning and predictions</i>		
d_b	integer	Broker symmetric pair-feature dimension, $d + d(d + 1)/2$.
$\psi_b(x_i, x_j)$	$\mathbb{R}^{d_b}$	Symmetric broker pair feature vector.
$\mathcal{H}_i^t = (\Xi_i^t, y_i^t, n_i^t)$	$\Xi_i^t \in \mathbb{R}^{d \times n_i^t}, y_i^t \in \mathbb{R}^{n_i^t}$	Agent i history; observation ℓ is partner agent type $\Xi_{i,\ell}^t$ and output $y_{i\ell}^t$.
$\mathcal{H}_b^t = (X_{b,1}^t, X_{b,2}^t, y_b^t, n_b^t)$	$X_{b,1}^t, X_{b,2}^t \in \mathbb{R}^{d \times n_b^t}, y_b^t \in \mathbb{R}^{n_b^t}$	Broker history; observation ℓ stores the two party types and output $y_{b\ell}^t$.
Θ_i^t	NN parameters	Agent i MLP weights.
Θ_b^t	NN parameters	Broker MLP weights.
$\mathcal{O}_i^t, \mathcal{O}_b^t$	Adam states	Persistent optimizer moments and step counters.
Δ_i^t, Δ_b^t	nonnegative integers	Observations added since the learner's previous training call.
$\mathbf{p}_i^t, \mathbf{p}_b^t$	integer sequences	Cumulative history counts at the end of learner initialization (period 0) and each later completed period.
$v_b^t(i, j)$	scalar	Symmetric broker prediction for unordered pair $\{i, j\}$.
$w_{j \leftarrow i}^t$	scalar	Receiver acceptance score.
$\bar{q}_i^t$	vector	Agent i 's partner-indexed empirical means.
C_i^t	vector	Agent i 's partner-indexed counts.
<i>Period state and market objects</i>		
s_i^t	$\mathbb{R}^2$	Satisfaction scores for self and broker channels.
χ_i^t	binary	Indicator that agent i has ever used the broker.
M_i^t	set	Agent i current-period counterparties.
z_i^t	channel	Channel chosen by demander i .
d_i^t	integer	Current-period demand.
D^t	subset of $[N]$	Current broker clients, $\{i : d_i^t > 0, z_i^t = \text{broker}\}$.
$\mathcal{A}_b^t$	subset of $[N]$	Broker access set, $\text{Roster}^t \cup D^t$.
Roster^t	subset of $[N]$	Standing roster.
rep_b^t	scalar	Broker reputation.
D_+^t	subset of $[N]$	Broker clients with positive residual offer quota.
$\mathcal{D}^t$	list	Demand records (i, z_i^t, d_i^t) with $d_i^t > 0$.
$\mathcal{R}^t$	list	Accepted records $(i, j, z, q, \hat{q}, o_1, o_2)$.
rng^t	RNG state	State of the simulation random-number stream.

The mutable simulation state at the end of period t is

$$S^t = \left(\vartheta, \text{rng}^t, \gamma, c, A, B, \bar{q}_{\text{cal}}, r, \phi, c_s, X^t, G^t, \left\{ \mathcal{H}_i^t, \Theta_i^t, \mathcal{O}_i^t, \Delta_i^t, \mathbf{p}_i^t, \bar{q}_i^t, C_i^t, s_i^t, \chi_i^t, M_i^t \right\}_{i=1}^N, \mathcal{H}_b^t, \Theta_b^t, \mathcal{O}_b^t, \Delta_b^t, \mathbf{p}_b^t, \text{Roster}^t, D^t, \text{rep}_b^t \right).$$

This tuple includes fixed realized model objects and all variables needed for later transitions. Implementation-only scratch buffers and diagnostic-only counters are omitted.

The matching function is

$$f(x_i, x_j) = \rho \frac{x_i^\top c + x_j^\top c}{2} + (1 - \rho) g(x_i, x_j) x_i^\top A x_j, \quad g(x_i, x_j) = 1 + \delta \operatorname{sign}(x_i^\top B x_j).$$

2 Top-Level Simulation

Algorithm 1 Brokerage Simulation

Require: Parameters ϑ and seed.

Ensure: Final state S^T and **measure table** $\mathcal{M} \in \mathbb{R}^{T \times m}$.

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1:  $S^0 \leftarrow \text{INITIALIZE}(\vartheta, \text{seed})$ 
2: for  $t = 1, \dots, T$  do
3:    $(S^t, \mathcal{R}^t, Y^t) \leftarrow \text{PERIODUPDATE}(S^{t-1}, t, \vartheta)$ 
4:    $\mathcal{M}_t \leftarrow Y^t$ 
5: return  $(S^T, \mathcal{M})$ 
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3 Training

For any one-hidden-layer predictor with input dimension p_{in} and hidden width h ,

$$\text{NN}_\Theta(\xi) = w_2^\top \text{ReLU}(W_1 \xi + b_1) + b_2, \quad \Theta = (W_1, b_1, w_2, b_2),$$

where $\xi \in \mathbb{R}^{p_{\text{in}}}$, $W_1 \in \mathbb{R}^{h \times p_{\text{in}}}$, $b_1, w_2 \in \mathbb{R}^h$, and $b_2 \in \mathbb{R}$.

$$d_b = d + \frac{d(d+1)}{2}, \quad \psi_b(x_i, x_j) = \left[x_i + x_j; \operatorname{vech}\left(\frac{x_i x_j^\top + x_j x_i^\top}{2}\right) \right].$$

Agent networks use $(p_{\text{in}}, h) = (d, 2d)$; the broker network uses $(p_{\text{in}}, h) = (d_b, 8d)$.

$$\psi_b(x_i, x_j) = \psi_b(x_j, x_i)$$

so broker histories produce one training column per observed unordered pair.

Local variables for training

Symbol	Dimension/domain	Definition
p_{in}, h	integers	Predictor input dimension and hidden width.
ξ	$\mathbb{R}^{p_{\text{in}}}$	Predictor input vector.
Θ	NN parameters	Generic MLP parameters (W_1, b_1, w_2, b_2) .
W_1, b_1, w_2, b_2	matrix, vectors, scalar	One-hidden-layer MLP weights and biases.
d_b	integer	Broker symmetric pair-feature dimension.
ψ_b	function	Symmetric broker pair-feature map.
$\Xi_{\text{tr}}, y_{\text{tr}}$	$\mathbb{R}^{p_{\text{in}} \times n}, \mathbb{R}^n$	Generic training inputs and targets.
$\mathcal{L}$	scalar	Full-batch MSE objective.
e	integer	Update-step index.
(m, v, τ)	Adam state	Persistent Adam moments and step counter; $\mathcal{O}_i, \mathcal{O}_b$ are the agent and broker instances.
$\beta_1, \beta_2, \epsilon$	scalars	Adam constants $(0.9, 0.999, 10^{-8})$.

Local variables for training (continued)

Symbol	Dimension/domain	Definition
n_i, n_b	integers	Current values of history lengths n_i^t and n_b^t .
L_i, L_b	ordered index sets	Training-window indices from the most recent W_p completed simulation periods, with period 0 included only until W_p later periods exist; subsampled evenly to at most M .
Δ_i, Δ_b	integers	New observations since last training call.
E_i, E_b	integers	Agent and broker update-step counts.
$\Xi_{i,\text{tr}}, y_{i,\text{tr}}$	$\mathbb{R}^{d \times L_i }, \mathbb{R}^{ L_i }$	Agent i training view of $\mathcal{H}_i^t$.
$\Psi_{b,\text{tr}}, y_{b,\text{tr}}$	$\mathbb{R}^{d_b \times L_b }, \mathbb{R}^{ L_b }$	Broker symmetric-feature training view of $\mathcal{H}_b^t$.

Algorithm 2 TrainNN

Require: Network parameters Θ , inputs $\Xi_{\text{tr}} \in \mathbb{R}^{p_{\text{in}} \times n}$, targets $y_{\text{tr}} \in \mathbb{R}^n$, steps E , learning rate η_{lr} , persistent Adam state (m, v, τ) .

- 1: **for** $e = 1, \dots, E$ **do**
 - 2: $g \leftarrow \nabla_{\Theta} \mathcal{L}(\Theta)$, $\mathcal{L}(\Theta) = n^{-1} \sum_{\ell=1}^n (\text{NN}_{\Theta}(\Xi_{\text{tr}, \ell}) - y_{\text{tr}, \ell})^2$
 - 3: $\tau \leftarrow \tau + 1$
 - 4: $m \leftarrow \beta_1 m + (1 - \beta_1)g$, $v \leftarrow \beta_2 v + (1 - \beta_2)g \odot g$
 - 5: $\Theta \leftarrow \Theta - \eta_{\text{lr}} \frac{m/(1 - \beta_1^{\tau})}{\sqrt{v/(1 - \beta_2^{\tau})} + \epsilon}$
 - 6: **return** $\Theta, (m, v, \tau)$
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Within period-level routines, state variables without a superscript refer to their current-period values.

Algorithm 3 TrainPredictors

Require: State S , period t , parameters $(E_{\text{init}}, E_{\text{steps}}, \eta_{\text{lr}}, W_p, M)$.

- 1: **for** $i = 1, \dots, N$ in parallel **do**
 - 2: $n_i \leftarrow n_i^t$, $\Delta_i \leftarrow$ observations added to $\mathcal{H}_i$ since its last training call.
 - 3: **if** $n_i > 0$, $\Delta_i > 0$, and $i \bmod 2 = t \bmod 2$ **then**
 - 4: Let $P_i \leftarrow |\mathbf{p}_i|$ and $b_i \leftarrow 0$ if $P_i \leq W_p$, otherwise $b_i \leftarrow (\mathbf{p}_i)_{P_i - W_p}$; set $L_i \leftarrow \{b_i + 1, \dots, n_i\}$.
If $|L_i| > M$, keep M spaced indices across L_i , including its first and last indices (or only its last index when $M = 1$).
 - 5: $\Xi_{i,\text{tr}} \leftarrow (\Xi_i^t)_{\cdot L_i}$, $y_{i,\text{tr}} \leftarrow (y_i^t)_{L_i}$.
 - 6: $E_i \leftarrow \max\{E_{\text{steps}}, \lceil E_{\text{init}} \Delta_i / n_i \rceil\}$.
 - 7: $(\Theta_i, \mathcal{O}_i) \leftarrow \text{TRAINNN}(\Theta_i, \Xi_{i,\text{tr}}, y_{i,\text{tr}}, E_i, \eta_{\text{lr}}, \mathcal{O}_i)$; $\Delta_i \leftarrow 0$.
 - 8: $n_b \leftarrow n_b^t$, $\Delta_b \leftarrow$ broker observations since last training.
 - 9: **if** $n_b > 0$ and $\Delta_b > 0$ **then**
 - 10: Let $P_b \leftarrow |\mathbf{p}_b|$ and $b_b \leftarrow 0$ if $P_b \leq W_p$, otherwise $b_b \leftarrow (\mathbf{p}_b)_{P_b - W_p}$; set $L_b \leftarrow \{b_b + 1, \dots, n_b\}$.
If $|L_b| > M$, keep M spaced indices across L_b , including its first and last indices (or only its last index when $M = 1$).
 - 11: $\Psi_{b,\text{tr}} \leftarrow \left[\psi_b \left((X_{b,1}^t)_{\cdot \ell}, (X_{b,2}^t)_{\cdot \ell} \right) \right]_{\ell \in L_b}$.
 - 12: $y_{b,\text{tr}} \leftarrow (y_b^t)_{L_b}$.
 - 13: $E_b \leftarrow \max\{E_{\text{steps}}, \lceil E_{\text{init}} \Delta_b / n_b \rceil\}$.
 - 14: $(\Theta_b, \mathcal{O}_b) \leftarrow \text{TRAINNN}(\Theta_b, \Psi_{b,\text{tr}}, y_{b,\text{tr}}, E_b, \eta_{\text{lr}}, \mathcal{O}_b)$; $\Delta_b \leftarrow 0$.
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4 Initialization

Calibration draws random pairs (i_ℓ, j_ℓ) and sets

$$\bar{q}_{\text{cal}} = \frac{1}{10,000} \sum_{\ell=1}^{10,000} \{Q + f(x_{i_\ell}, x_{j_\ell})\},$$

$$r = f_r \bar{q}_{\text{cal}}, \quad \phi = c_s = \lambda_c \bar{q}_{\text{cal}}.$$

Local variables for Initialize

Symbol	Dimension/domain	Definition
λ_c	scalar	Cost rate used to set ϕ and c_s .
f_r	scalar	Reservation coefficient; sets $r = f_r \bar{q}_{\text{cal}}$ (default 0.60).
(i_ℓ, j_ℓ)	pair in $[N]^2$	Random calibration pair.
d_γ	integer	Number of active curve dimensions.
p_{rewire}	scalar	Watts-Strogatz rewiring probability.
ν_ℓ	integer	Frequency for active curve coordinate ℓ .
ψ_ℓ	scalar	Phase for active curve coordinate ℓ .
A_0	$\mathbb{R}^{d \times d}$	Raw matrix used to construct A .
Σ_x	$\mathbb{R}^{d \times d}$	Empirical agent type second moment.
B_0	$\mathbb{R}^{d \times d}$	Raw zero-trace symmetric regime matrix.
B_{raw}	$\mathbb{R}^{d \times d}$	Orthogonalized pre-normalization regime matrix.
$\Psi_{b, \text{seed}}, y_{b, \text{seed}}$	$\mathbb{R}^{d_b \times n_b^0}, \mathbb{R}^{n_b^0}$	Broker seed training batch using symmetric pair features.

Local variables for InitializeAgent

Symbol	Dimension/domain	Definition
σ_x	scalar	Agent type noise scale.
u_i	scalar	Agent i curve position.
ϵ_i	$\mathbb{R}^d$	Agent type noise for agent i .
π_{ij}^{ent}	scalar	Entrant attachment probability.
$\mathcal{V}_i^{\text{ent}}$	subset of $[N] \setminus \{i\}$	Entrant attachment set.

Algorithm 4 InitializeAgent

Require: State S , agent slot i , period counter t , parameters ϑ .

Ensure: Agent slot i is initialized in S .

- 1: Draw $u_i \sim \text{Unif}[0, 1]$ and $\epsilon_i \sim \mathcal{N}(0, \sigma_x^2 I_d/d)$.
- 2: $x_i^t \leftarrow (\gamma(u_i) + \epsilon_i) / \|\gamma(u_i) + \epsilon_i\|_2$.
- 3: Reset $\mathcal{H}_i^t = (\Xi_i^t, y_i^t, n_i^t)$ to zero observations, $\bar{q}_i^t$, C_i^t , χ_i^t , M_i^t , and Δ_i^t . Draw W_{1i}^t with He initialization and set $b_{1i}^t = 0$, $w_{2i}^t = 0$, $b_{2i}^t = Q$; reset $\mathcal{O}_i^t$ to zero.
- 4: **if** $t = 0$ **then**
- 5: $s_i^t \leftarrow (0, 0)$.
- 6: **else**
- 7: Remove i from Roster^t , D^t , and M_j^t for all j ; delete all incident edges in G^t ; set $\bar{q}_{ji}^t, C_{ji}^t \leftarrow 0$ for all $j \neq i$.

8: Sample $\mathcal{V}_i^{\text{ent}} \subset [N] \setminus \{i\}$ without replacement, $|\mathcal{V}_i^{\text{ent}}| = \min(\lfloor k_G/2 \rfloor, N-1)$, with

$$\pi_{ij}^{\text{ent}} \propto \exp\{-\|x_i^t - x_j^t\|_2^2\}.$$

9: Add edges (i, j) to G^t for all $j \in \mathcal{V}_i^{\text{ent}}$.

10: $s_{i,\text{self}}^t \leftarrow \begin{cases} |\mathcal{V}_i^{\text{ent}}|^{-1} \sum_{j \in \mathcal{V}_i^{\text{ent}}} s_{j,\text{self}}^t, & \mathcal{V}_i^{\text{ent}} \neq \emptyset, \\ 0, & \mathcal{V}_i^{\text{ent}} = \emptyset. \end{cases}$

11: $s_{i,\text{broker}}^t \leftarrow \text{rep}_b^t$.

12: Set $\mathbf{p}_i^t \leftarrow (0)$, treating entrant initialization as period 0 of its tenure.

Algorithm 5 Initialize

Require: Parameters ϑ and seed.

Ensure: Initial state S^0 .

- 1: Draw curve frequencies $\nu_\ell \sim \text{Unif}\{1, \dots, 5\}$ and phases $\psi_\ell \sim \text{Unif}[0, 2\pi)$ for $\ell = 1, \dots, d_\gamma$, defining γ .
- 2: Allocate empty agent and broker containers in S^0 and store γ .
- 3: **for** $i = 1, \dots, N$ **do**
- 4: INITIALIZEAGENT($S^0, i, 0, \vartheta$).
- 5: Draw $u_c \sim \text{Unif}[0, 1]$ and $\epsilon_c \sim \mathcal{N}(0, \sigma_x^2 I_d/d)$; set $c \leftarrow \gamma(u_c) + \epsilon_c$ without re-normalizing.
- 6: Draw $A_0 \in \mathbb{R}^{d \times d}$ iid standard normal and set $A \leftarrow A_0^\top A_0 \cdot d / \text{tr}(A_0^\top A_0)$.
- 7: $\Sigma_x \leftarrow N^{-1} \sum_i x_i^0 (x_i^0)^\top$.
- 8: Draw symmetric zero-trace B_0 and set

$$B_{\text{raw}} \leftarrow B_0 - \frac{\text{tr}(\Sigma_x B_0 \Sigma_x A)}{\text{tr}(\Sigma_x A \Sigma_x A)} A, \quad B \leftarrow B_{\text{raw}} / \|B_{\text{raw}}\|_F.$$

- 9: Calibrate $(\bar{q}_{\text{cal}}, r, \phi, c_s)$.
- 10: Build G^0 as a Watts-Strogatz graph on $[N]$ with degree k_G and rewiring probability p_{rewire} ; add isolated broker node $N+1$.
- 11: Initialize $\mathcal{H}_b^0$ and set W_{1b}^0 using He initialization, $b_{1b}^0 = 0$, $w_{2b}^0 = 0$, $b_{2b}^0 = Q$, $\mathcal{O}_b^0 = 0$, and $\text{rep}_b^0 = 0$.
- 12: Sample $\text{Roster}^0 \subset [N]$ uniformly with $|\text{Roster}^0| = \lceil \alpha_R N \rceil$; add broker edges $(i, N+1)$ for all roster members.
- 13: **for** each unordered non-broker edge $\{i, j\} \in \mathcal{E}_G^0$ **do**
- 14: Draw q_{ij} once; append (x_j^0, q_{ij}) to $\mathcal{H}_i^0$ and (x_i^0, q_{ij}) to $\mathcal{H}_j^0$; update $\bar{q}_{ij}^0, C_{ij}^0, \bar{q}_{ji}^0, C_{ji}^0$.
- 15: Let $\mathcal{B}_{\text{seed}} \leftarrow \{\{i, j\} \in \mathcal{E}_G^0 : i, j \in \text{Roster}^0\}$; shuffle and keep at most 100 edges.
- 16: **for** each $\{i, j\} \in \mathcal{B}_{\text{seed}}$ **do**
- 17: Append (x_i^0, x_j^0, q_{ij}) to $\mathcal{H}_b^0$; increment n_b^0 .
- 18: **if** $n_b^0 > 0$ **then**
- 19: Set rep_b^0 to mean broker seed output.
- 20: **else**
- 21: Set $\text{rep}_b^0 \leftarrow 0$.
- 22: Set $s_{i,\text{self}}^0$ to mean own seed output and $s_{i,\text{broker}}^0 \leftarrow \text{rep}_b^0$.
- 23: Set $\mathbf{p}_i^0 \leftarrow (n_i^0)$ for every agent and $\mathbf{p}_b^0 \leftarrow (n_b^0)$, so initialization is period 0; set $\Delta_i^0 \leftarrow n_i^0$ and $\Delta_b^0 \leftarrow n_b^0$.
- 24: **for** each i with $n_i^0 > 0$ **do**
- 25: $(\Theta_i^0, \mathcal{O}_i^0) \leftarrow \text{TRAINNN}(\Theta_i^0, \Xi_i^0, y_i^0, E_{\text{init}}, \eta_{\text{lr}}, \mathcal{O}_i^0)$; $\Delta_i^0 \leftarrow 0$.

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26: if  $n_b^0 > 0$  then
27:    $\Psi_{b,\text{seed}} \leftarrow \left[ \psi_b \left( (X_{b,1}^0)_{\cdot \ell}, (X_{b,2}^0)_{\cdot \ell} \right) \right]_{\ell=1}^{n_b^0}, y_{b,\text{seed}} \leftarrow y_b^0.$ 
28:    $(\Theta_b^0, \mathcal{O}_b^0) \leftarrow \text{TRAINNN}(\Theta_b^0, \Psi_{b,\text{seed}}, y_{b,\text{seed}}, E_{\text{init}}, \eta_{\text{lr}}, \mathcal{O}_b^0); \Delta_b^0 \leftarrow 0.$ 
29: return  $S^0.$ 

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5 Channel Choice

Local variables for ChooseChannel

Symbol	Dimension/domain	Definition
U_{self}	scalar	Self-channel score.
U_{broker}	scalar	Broker-channel score.

Algorithm 6 ChooseChannel

Require: Agent i state (s_i, χ_i) , broker reputation rep_b , RNG.

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1:  $U_{\text{self}} \leftarrow s_{i,\text{self}}.$ 
2:  $U_{\text{broker}} \leftarrow \chi_i s_{i,\text{broker}} + (1 - \chi_i) \text{rep}_b.$ 
3: if  $U_{\text{self}} > U_{\text{broker}}$  then
4:   return self.
5: if  $U_{\text{broker}} > U_{\text{self}}$  then
6:   return broker.
7: return self w.p. 1/2, else broker  $\triangleright U_{\text{self}} = U_{\text{broker}}.$ 

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6 One Period

Local variables for PeriodUpdate

Symbol	Dimension/domain	Definition
K	integer	Active-demand bound.
p_{demand}	scalar	Per-position demand probability.
α_R	scalar	Standing broker roster share.
p_{roster}	scalar	Roster-churn probability.
η_{exit}	scalar	Agent exit probability.
Δ_{net}	integer	Network-measure interval.

Algorithm 7 PeriodUpdate

Require: Previous state S^{t-1} , period t , parameters ϑ .

Ensure: Updated state S^t , accepted records $\mathcal{R}^t$, and **pre-turnover measure row** Y^t .

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1: Clear accumulators, set  $M_i^t \leftarrow \emptyset$  for all  $i$ , and set  $D^t \leftarrow \emptyset.$ 
2: Refresh roster: remove each roster member with probability  $p_{\text{roster}}$ , then replenish uniformly to
   size  $\lceil \alpha_R N \rceil.$ 
3:  $\mathcal{E}_G^t \leftarrow (\mathcal{E}_G^t \setminus \{\{i, N+1\} : i \in [N]\}) \cup \{\{i, N+1\} : i \in \text{Roster}^t\}.$ 
4:  $\mathcal{D}^t \leftarrow \emptyset.$ 
5: for  $i = 1, \dots, N$  do

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6:   Draw  $d_i^t \sim \text{Binomial}(K, p_{\text{demand}})$ .
7:   if  $d_i^t > 0$  then
8:      $z_i^t \leftarrow \text{CHOOSECHANNEL}(i, \text{rep}_b^{t-1})$ .
9:     Append  $(i, z_i^t, d_i^t)$  to  $\mathcal{D}^t$ .
10:    if  $z_i^t = \text{broker}$  then
11:      Add  $i$  to  $D^t$ .
12:   $\mathcal{E}_G^t \leftarrow (\mathcal{E}_G^t \setminus \{\{i, N+1\} : i \in [N]\}) \cup \{\{i, N+1\} : i \in \text{Roster}^t \cup D^t\}$ .
13:   $\text{TRAINPREDICTORS}(S, t, \vartheta)$ .
14:   $\mathcal{R}^t \leftarrow \text{OFFERMARKET}(S, \mathcal{D}^t, \vartheta)$ .
15:  Synchronize broker-node edges to  $\text{Roster}^t$ ,  $D^t$ , and current broker-channel matches.
16:   $\text{UPDATESATISFACTION}(S, \mathcal{D}^t, \mathcal{R}^t)$ .
17:  if  $D^t \neq \emptyset$  then
18:     $\text{rep}_b^t \leftarrow |D^t|^{-1} \sum_{i \in D^t} s_{i, \text{broker}}^t$ .
19:  else
20:     $\text{rep}_b^t \leftarrow \text{rep}_b^{t-1}$ .
21:  if  $t \bmod \Delta_{\text{net}} = 0$  then
22:    Recompute broker betweenness, Burt constraint, and effective size on the post-matching,
    pre-turnover graph  $G^t$ .
23:   $Y^t \leftarrow \text{COLLECTMEASURES}(S^t, \mathcal{R}^t, \vartheta)$  on the post-matching, pre-turnover state.
24:  Append  $n_i^t$  to  $\mathbf{p}_i^t$  for every agent and append  $n_b^t$  to  $\mathbf{p}_b^t$ , recording the end-of-period training-
    window boundaries.
25:  for  $i = 1, \dots, N$  do
26:    if agent  $i$  exits with probability  $\eta_{\text{exit}}$  then
27:       $\text{INITIALIZEAGENT}(S^t, i, t, \vartheta)$ .
28:  Synchronize broker-node edges to the post-turnover roster, clients, and active matches.
29:  return  $(S^t, \mathcal{R}^t, Y^t)$ .

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7 Offer Market

Local variables for OfferMarket

Symbol	Dimension/domain	Definition
$\hat{q}_i^t$	$\mathbb{R}^d \rightarrow \mathbb{R}$	Agent predictor.
$\hat{q}_b^t$	$\mathbb{R}^{d_b} \rightarrow \mathbb{R}$	Broker predictor.
$\mathcal{N}_G^t(i)$	subset of $[N]$	Non-broker graph neighbors of i .
U_i^t	subset of $[N]$	Self-search stranger pool independently sampled for demander i .
$n_{\text{strangers}}$	integer	Per-demander stranger-pool size.
$v_{i \rightarrow j}^{\text{self}, t}$	scalar	Self-offer score.
Ω_i	integer	Residual offer quota for demander i .
Ω	$\mathbb{Z}_{\geq 0}^N$	Residual quota vector.
O^t	map	Directed offer book.
O_{ij}^t	offer record	Record (i, j, z, v) for offer $i \rightarrow j$.
$\mathcal{P}_O^t$	set	Unordered pairs represented in O^t .
$\mathcal{U}_i^{\text{self}}$	subset of $[N]$	Self-search candidate set.
$\mathcal{I}_i^{\text{self}}$	list	Self-search candidates sorted by score.
$\mathcal{P}_b^t$	set	Feasible unordered broker-pair set.

Local variables for OfferMarket (continued)

Symbol	Dimension/domain	Definition
$\mathcal{I}_{b,i}$	list	Accessible broker candidates for client i , sorted by v_b^t .
$\mathcal{B}_b^t$	list	Selected directed broker offers.
o, o', o_{ij}, o_{ji}	offer records	Offer records; fields include $o.z$ and $o.from$.
z^*	channel	Accepted-record channel.
α, β	indices	Ordered endpoints induced by one unordered broker pair.
ι	$\{0, 1\}$	Return value of AddOffer.

The symmetric broker score is

$$v_b^t(i, j) = \hat{q}_b^t(\psi_b(x_i^t, x_j^t)).$$

Let $\mathcal{N}_G^t(i) = \{j \in [N] : (i, j) \in \mathcal{E}_G^t\}$ exclude the broker node. The self-search proposal score is defined for every graph neighbor and for the non-neighbors in the demander-specific stranger pool:

$$v_{i \rightarrow j}^{\text{self}, t} = \begin{cases} \bar{q}_{ij}^t, & j \in \mathcal{N}_G^t(i), C_{ij}^t > 0, \\ \hat{q}_i^t(x_j^t), & j \in \mathcal{N}_G^t(i), C_{ij}^t = 0, \\ \hat{q}_i^t(x_j^t), & j \in U_i^t \setminus \mathcal{N}_G^t(i), \\ \perp, & \text{otherwise.} \end{cases}$$

Thus direct pair means replace predictor extrapolations when pair-specific experience exists; $\perp$ denotes no valid self-search score. The receiver-side acceptance score for a one-sided offer is

$$w_{j \leftarrow i}^t = \begin{cases} \bar{q}_{ji}^t, & (i, j) \in \mathcal{E}_G^t, C_{ji}^t > 0, \\ \hat{q}_j^t(x_i^t), & \text{otherwise.} \end{cases}$$

The market creates at most d_i^t outgoing offers from demander i . Receivers have no per-period capacity constraint; acceptance is pairwise, so a receiver may accept multiple offers in the same period. A demanded position can remain unmatched if no acceptable offer is formed.

Algorithm 8 AddOffer

Require: Offer book O^t , unordered-pair set $\mathcal{P}_O^t$, offer (i, j, z, v) .

- 1: **if** O_{ij}^t is defined **then**
 - 2: **return** 0.
 - 3: $O_{ij}^t \leftarrow (i, j, z, v)$.
 - 4: **if** O_{ji}^t is undefined **then**
 - 5: $\mathcal{P}_O^t \leftarrow \mathcal{P}_O^t \cup \{\{i, j\}\}$.
 - 6: **return** 1.
-

Algorithm 9 OfferMarket

Require: State S , demand records $\mathcal{D}^t$, parameters ϑ .

Ensure: Accepted records $\mathcal{R}^t$.

- 1: For all $(i, z_i, d_i) \in \mathcal{D}^t$, set residual offer quota $\Omega_i \leftarrow d_i$.
- 2: $O^t \leftarrow \emptyset$, $\mathcal{P}_O^t \leftarrow \emptyset$, $\mathcal{R}^t \leftarrow \emptyset$.
- 3: **for** each $(i, \text{self}, d_i) \in \mathcal{D}^t$ with $\Omega_i > 0$ **do**

```

4:   Draw  $U_i^t \subset [N]$  uniformly without replacement,  $|U_i^t| = \min(n_{\text{strangers}}, N)$ .
5:    $\mathcal{U}_i^{\text{self}} \leftarrow \mathcal{N}_G^t(i) \cup (U_i^t \setminus \mathcal{N}_G^t(i))$ .
6:   Remove from  $\mathcal{U}_i^{\text{self}}$  the elements of  $\{i\} \cup M_i^t$ .
7:    $\mathcal{I}_i^{\text{self}} \leftarrow \text{argsort}_{j \in \mathcal{U}_i^{\text{self}}: v_{i \rightarrow j}^{\text{self}, t} > r}(-v_{i \rightarrow j}^{\text{self}, t})$ .
8:   for the first  $\min(\Omega_i, |\mathcal{I}_i^{\text{self}}|)$  agents  $j$  in  $\mathcal{I}_i^{\text{self}}$  do
9:        $\text{ADDOFFER}(O^t, \mathcal{P}_O^t, i, j, \text{self}, v_{i \rightarrow j}^{\text{self}, t})$ .
10:  $\mathcal{A}_b^t \leftarrow \text{Roster}^t \cup D^t$ .
11:  $D_+^t \leftarrow \{i : (i, \text{broker}, d_i) \in \mathcal{D}^t, \Omega_i > 0\}$ .
12:  $\mathcal{P}_b^t \leftarrow \{\{i, j\} : i \neq j, (i \in D_+^t, j \in \mathcal{A}_b^t) \text{ or } (j \in D_+^t, i \in \mathcal{A}_b^t)\}$ .
13:  $\mathcal{B}_b^t \leftarrow \emptyset$ .
14: for each  $i \in D_+^t$  do
15:      $\mathcal{I}_{b,i} \leftarrow \text{argsort}_{j \in \mathcal{A}_b^t: \{i, j\} \in \mathcal{P}_b^t, v_b^t(i, j) > r}(-v_b^t(i, j))$ .
16:     for the first  $\min(\Omega_i, |\mathcal{I}_{b,i}|)$  agents  $j$  in  $\mathcal{I}_{b,i}$  do
17:         Append  $(i, j, v_b^t(i, j))$  to  $\mathcal{B}_b^t$ .
18: Sort  $\mathcal{B}_b^t$  by descending  $v_b^t$  with the deterministic unordered-pair tie key.
19: for each  $(i, j, v) \in \mathcal{B}_b^t$  do
20:     if  $\Omega_i > 0$  then
21:          $\iota \leftarrow \text{ADDOFFER}(O^t, \mathcal{P}_O^t, i, j, \text{broker}, v)$ .
22:          $\Omega_i \leftarrow \Omega_i - \iota$ .
23: for each  $\{i, j\} \in \mathcal{P}_O^t$  do
24:      $o_{ij} \leftarrow O_{ij}^t$  if defined, else  $\emptyset$ ;  $o_{ji} \leftarrow O_{ji}^t$  if defined, else  $\emptyset$ .
25:     if  $o_{ij} \neq \emptyset$  and  $o_{ji} \neq \emptyset$  then
26:          $\text{FINALIZEMATCH}(S, \mathcal{R}^t, o_{ij}, o_{ji})$ .
27:     else if  $o_{ij} \neq \emptyset$  and  $w_{j \leftarrow i}^t > r$  then
28:          $\text{FINALIZEMATCH}(S, \mathcal{R}^t, o_{ij}, \emptyset)$ .
29:     else if  $o_{ji} \neq \emptyset$  and  $w_{i \leftarrow j}^t > r$  then
30:          $\text{FINALIZEMATCH}(S, \mathcal{R}^t, o_{ji}, \emptyset)$ .
31: return  $\mathcal{R}^t$ .

```

Algorithm 10 FinalizeMatch

Require: State S , accepted-record list $\mathcal{R}^t$, offer $o = (i, j, z, v)$, optional reverse offer o' .

```

1: if  $j \in M_i^t$  then
2:     return .
3: Draw  $q_{ij}^t = Q + f(x_i^t, x_j^t) + \varepsilon$ ,  $\varepsilon \sim \mathcal{N}(0, \sigma_\varepsilon^2)$ .
4: Append columns  $x_j^t$  to  $\Xi_i$  and  $x_i^t$  to  $\Xi_j$ ; append  $q_{ij}^t$  to  $y_i$  and  $y_j$ ; increment  $n_i$  and  $n_j$ .
5: Update  $\bar{q}_{ij}$ ,  $C_{ij}$  and  $\bar{q}_{ji}$ ,  $C_{ji}$ .
6: Add edge  $(i, j)$  to  $G^t$ ; add  $j$  to  $M_i^t$  and  $i$  to  $M_j^t$ .
7: if  $z = \text{broker}$  or  $(o' \neq \emptyset \text{ and } o'.z = \text{broker})$  then
8:     Append the two party types  $x_i^t$  and  $x_j^t$  to  $X_{b,1}$  and  $X_{b,2}$ ; append  $q_{ij}^t$  to  $y_b$ ; increment  $n_b$ .
9:      $z^* \leftarrow \text{broker}$ .
10: else
11:      $z^* \leftarrow \text{self}$ .
12: Append accepted record  $(i, j, z^*, q_{ij}^t, v, o, o')$  to  $\mathcal{R}^t$ .

```

8 Satisfaction and Measures

Local variables for satisfaction

Symbol	Dimension/domain	Definition
Y_i	scalar	Agent i period satisfaction numerator.
$n_{i,b}$	integer	Accepted broker offers credited to demander i .
$\tilde{q}_i^t$	scalar	Channel-specific satisfaction input.
ζ	accepted record	Iterator over $\mathcal{R}^t$.
$\mathbf{1}\{\cdot\}$	indicator	Equals 1 when its condition is true and 0 otherwise.

Algorithm 11 UpdateSatisfaction

Require: State S , demand records $\mathcal{D}^t$, accepted records $\mathcal{R}^t$.

- 1: **for** each $(i, z_i, d_i) \in \mathcal{D}^t$ **do**
 - 2: $Y_i \leftarrow \sum_{\zeta \in \mathcal{R}^t} \sum_{o \in \{\zeta.o_1, \zeta.o_2\}} \mathbf{1}\{o \neq \emptyset, o.\text{from} = i\} \zeta.q.$
 - 3: $n_{i,b} \leftarrow \sum_{\zeta \in \mathcal{R}^t} \sum_{o \in \{\zeta.o_1, \zeta.o_2\}} \mathbf{1}\{o \neq \emptyset, o.\text{from} = i, o.z = \text{broker}\}.$
 - 4: **if** $z_i = \text{self}$ **then**
 - 5: $\tilde{q}_i^t \leftarrow Y_i/d_i - c_s.$
 - 6: **else**
 - 7: $\tilde{q}_i^t \leftarrow (Y_i - \phi n_{i,b})/d_i; \chi_i^t \leftarrow 1.$
 - 8: $s_{i,z_i}^t \leftarrow (1 - \omega)s_{i,z_i}^{t-1} + \omega\tilde{q}_i^t.$
-

CollectMeasures. $\text{COLLECTMEASURES}(S^t, \mathcal{R}^t, \vartheta)$ is a pure measure-collection function called after matching and before entry/exit turnover. It records demand, outsourcing, accepted matches, realized output, selected-sample prediction quality, deterministic holdout prediction quality, broker access and assessment counts, satisfaction, reputation, roster size, broker access size, agent-node degree summaries excluding the broker node, and cached broker network measures. The measures themselves do not feed back into S^{t+1} .